

you'll need to manually configure it, as HP's built-in solution abstracts much of the AI interaction layer.

GRAPHICS PERFORMANCE

Graphics are handled by the integrated AMD Radeon 860M, featuring 8 compute units clocked at 3 GHz. While this is not a gaming or workstation GPU, it's perfectly fine for light creative tasks like photo editing in Lightroom, 2D vector work, and even occasional 1080p video editing. More demanding workflows, especially those involving 3D modelling or 4K multi-layer timelines, will push the iGPU to its limits.

It's also worth noting that the GPU plays a supporting role in some of the AI workloads if the NPU is busy. This kind of dynamic compute resource management shows the strength of AMD's architecture, though it's mostly transparent to the end user.

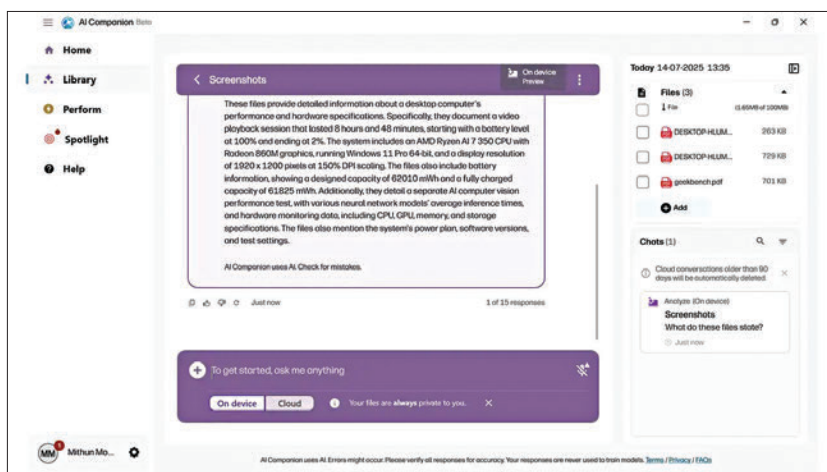
DISPLAY FOR PRODUCTIVITY

While not a high refresh rate panel, the 14-inch 1920 x 1200 display is well suited for productivity work. The 16:10 aspect ratio offers more vertical space, ideal for reading documents or writing code. Colour coverage stands at 100% sRGB and 83.3% DCI-P3, usable for semi-professional photo and video editing.

At 800 nits, brightness is sufficient for well-lit indoor spaces, though it does struggle under direct sunlight. We measured a maximum of 499 Nits on the panel. The display's matte finish helps reduce glare, and the panel shows no signs of PWM flicker or ghosting.

THERMALS AND NOISE

Thermal performance is reasonable but not exemplary. Under light to moderate loads, the EliteBook remains silent and cool. While idle, we saw temperatures usually hovering about 55 degrees Celsius. However, extended multi-core workloads (like batch rendering, long video calls with background blur, or simultaneous AI inference) begin to raise internal temperatures. The fan spins up audibly under sustained stress but remains within acceptable limits.



During normal use in an office environment with 24×7 air-conditioning, the unit spun up to 64 degrees Celsius which is a teensy bit higher than what we'd like to see. Surface temperatures peaked around 42 degrees.

HP's Smart Sense system does a decent job at managing thermal behaviour dynamically. It shifts between performance and eco modes based on application usage patterns. For example, the fan won't ramp up while watching a local video file or writing documents, but it will kick in during sustained browser-based tasks or Teams meetings.

The thermal design is a little conservative, to prioritise silence over peak performance. Users who need every bit of CPU potential might feel limited here.

BATTERY LIFE AND CHARGING

Battery endurance is another highlight. In our tests, the EliteBook 8 G1a ran for 8 hours and 48 minutes at 500 nits brightness while performing a mix of document editing, video playback, and browser-based tasks. Most folks run their monitors at 300-400 nits so you can easily expect much more battery life. That's a strong showing for a machine with a high-performance CPU, NPU, and relatively bright display.

Thanks to its 100W slim USB-C power adapter, charging is brisk. We consistently achieved over 50% charge in just 30 minutes, making it a good companion for airport lounges or between-meeting top-ups.

VERDICT

The HP EliteBook 8 G1a is an impressive showcase of what the future of business laptops looks like, one where AI is an integral part of the local computing experience. Its 50 TOPS NPU, AMD Ryzen AI processor, strong security framework, and intelligent battery management make it a strong contender in the AI PC category.

If HP can refine the pricing to be a bit more competitive, the EliteBook 8 G1a has the potential to lead the charge in AI-enabled mobile productivity. For now, it stands as a powerful, forward-looking laptop that's just a few steps away from being exceptional.

—Mithun Mohandas

SPECIFICATIONS

Processor: AMD Ryzen AI 7 350 (8-cores, 16-threads, 66 TOPS) | Graphics: AMD Radeon 860M GPU, 8 Compute Units | Memory: 32GB DDR5 5600MT/s (Supports 64 GB) | Storage: 1TB Western Digital PC NVMe (6000 Mb/s read, 5400 MB/s write) | Display: 14-inch WLED | Ports: 2 Thunderbolt 4, HDMI 2.1, Headphone/mic combo jack, 1 USB 3.2 Gen 2 Type-C, 1 USB 3.2 Gen 1 Type-A | Connectivity: MediaTek Wi-Fi 7, Bluetooth 5.4 | Dimensions: 31.5 x 22.2 x 1.55 cm | Weight: 1.39g | Battery: 3-cell, 62 Wh Li-ion polymer

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